Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

733
-22 -2.91%
Nov 21st, 2022

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

837
-19.00 -2.22%
Nov 21st, 2022

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

664
-4 -0.60%
Nov 21st, 2022

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

96.00
0.80 0.84%
Nov 21st, 2022

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

103.70
-0.60 -0.58%
Nov 21st, 2022

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

850
68 8.70%
Week Ending Nov 18th, 2022

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2301 (Jan) RMB/t (3pm close)

745.50
-8.00 -1.06%
Nov 21st, 2022 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
December 22 USD/dmt

95.80
-3.00 -3.04%
Nov 21st, 2022 (5.30 pm Print)

SHFE Rebar
RB2301 (Jan) RMB/t

3671
-45 -1.21%
Nov 21st, 2022 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

17.79
-0.20 -1.14%
Nov 18th, 2022

C5, W. Australia - Qingdao USD/t

7.83
-0.12 -1.45%
Nov 18th, 2022

Steel Price

Steel Rebar (China Domestic) RMB/t

3840
50 1.32%
Week Ending Nov 18th, 2022

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

130.20
1.47 1.14%
Week Ending Nov 18th, 2022

Steel Inventory in China
million tonnes

9.15
-0.58 -5.95%
Week Ending Nov 18th, 2022

Steel Price

Steel HRC (China Domestic) RMB/t

3860
50 1.31%
Week Ending Nov 18th, 2022

IRON ORE PORT STOCK INDEX (IOPI)

Nov 21st, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	733	-22	-2.9%	748	870	596	1083	95.17	-3.20	-3.3%	99.32	121.95	82.29	159.06
IOPI58	58% Fe Fines	664	-4	-0.6%	676	737	454	907	86.80	-0.75	-0.9%	90.38	103.47	64.78	134.15
IOPI65	65% Fe Fines	837	-19	-2.2%	881	1030	811	1265	109.21	-2.82	-2.5%	117.57	145.15	108.78	187.25

IRON ORE SEABORNE INDEX (IOSI)

Nov 21st, 2022		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	96.00	0.80	0.84%	100.53	124.37	83.90	159.45
IOSI65	65% Fe Fines	103.70	-0.60	-0.58%	111.34	143.66	94.45	185.82

MARKET COMMENTARY

The main contract closed 745.5 today. The traders' willingness to ship is general. The steel mills are purchase on demand. The overall trading sentiment of the market is general. PBF at Shandong port dealt 735 yuan/mt,decrease 5-15 yuan/mt; PBF at Tangshan port dealt 743-745 yuan/mt,derease 15-17 yuan/mt; SSF at Tangshan port dealt 615 yuan/mt. According to SMM tracking, global iron ore shipments last week totaled 30.98 million tons, up 4.7% month on month. Of which, 14.55 million tons were shipped from Australia to China, an increase of 18.1% month on month; Brazil sent 3.48 million tons to China, up 4.5% month on month. This week, SMM tracked that the total amount of Chinese iron ore arriving at the port was 25.96 million tons, up 3% month on month. However, due to the severe polluted weather in Hebei, Tangshan, Handan and Qinhuangdao continued to implement environmental restrictions on production, and some steel mills were also short of sinter supply to reduce hot metal output. Superimposed by the fact that Jiangsu Province continues to implement the crude steel reduction, it is expected that the hot metal production will decline slightly this week, and the port inventory may accumulate again. In addition, the news that India has

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Nov 18th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	850	68	8.7%	866	1062	730	1405	107.20	10.97	11.40%	110.85	143.67	95.00	199.13

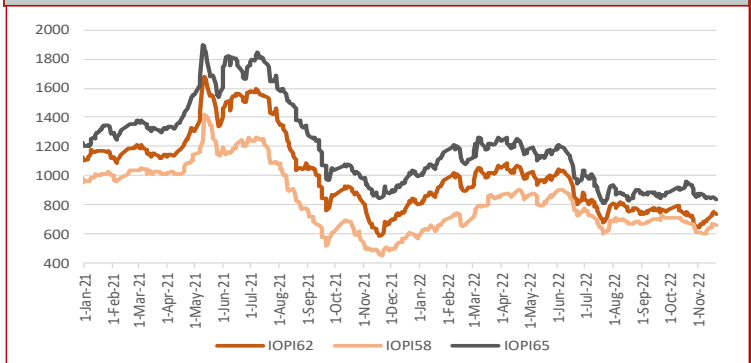
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Nov 18th, 2022				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	869	5.6%	779	1645	122.92	7.83%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	910	4.6%	780	1630	128.72	6.81%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	705	0.0%	620	1310	99.73	2.12%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	892	5.4%	800	1752	126.18	7.67%	117.19	272.32

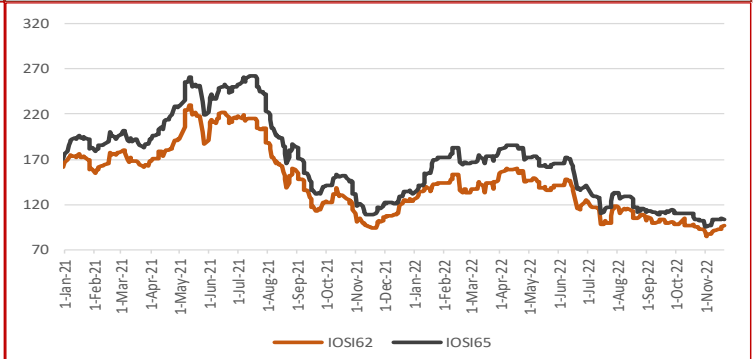
Week Ending Nov 18th, 2022	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	801.27	1.44%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =7.06942

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Nov 21st, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	July	August	September	October	MTD	QTD	YTD	June	August	September	October	MTD	QTD	YTD
IOPI62	62% Fe Fines	770	781	759	736	748	753	870	106.01	106.64	101.14	95.56	99.32	100.78	121.95
IOPI58	58% Fe Fines	682	685	697	683	676	678	737	94.43	94.05	93.56	89.34	90.38	91.26	103.47
IOPI65	65% Fe Fines	915	878	875	918	881	889	1030	126.73	120.31	117.19	120.08	117.57	119.58	145.15

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Nov 21st, 2022		CFR Qingdao, USD/dry tonne						
Index	Fe Content	July	August	September	October	MTD	QTD	YTD
IOSI62	62% Fe Fines	110.07	110.64	101.12	95.80	100.53	102.52	124.37
IOSI65	65% Fe Fines	124.60	112.53	111.57	106.70	111.34	114.11	143.66

FREIGHT RATES

Nov 18th, 2022		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	7.83	-0.12	-1.45%	3.57	16.77
Tubarao - Qingdao	C3	17.79	-0.20	-1.14%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Nov 18th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	July	August	September	October	MTD	QTD	YTD	July	August	September	October	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	906	900	873	877	866	876	1062	120.36	118.50	112.75	109.81	110.85	113.22	143.67

IRON ORE INDEX PREMIUMS/DISCOUNTS

Nov 21st, 2022		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-69	-9.41%
IOPI65	65% Fe Fines	104	14.19%

Nov 21st, 2022		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	7.70	8.02%

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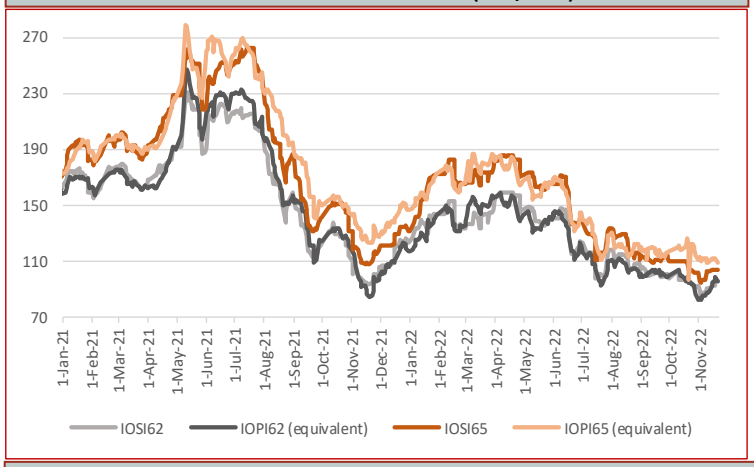
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Nov 21st, 2022	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	676	-22	-57
SIMEC Fines	602	-23	-131
PB Fines	703	-22	-30
Newman Fines	701	-22	-32
MAC Fines	682	-23	-51
Jimblebar Blended Fines	597	-22	-136
Carajas Fines	913	-22	180
Brazilian SSF	694	-23	-39
Brazilian Blend Fines	716	-22	-17
RTX Fines	615	-22	-118
West Pilbara Fines	645	-22	-88

Nov 21st, 2022	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	91.50	0.80	-4.50
SIMEC Fines	88.00	0.80	-8.00
PB Fines	92.25	0.80	-3.75
Newman Fines	95.15	0.80	-0.85
MAC Fines	92.25	0.80	-3.75
Jimblebar Blended Fines	84.65	0.80	-11.35
Carajas Fines	125.55	0.80	29.55
Brazilian SSF	99.75	0.80	3.75
Brazilian Blend Fines	101.40	0.80	5.40
RTX Fines	85.90	0.80	-10.10
West Pilbara Fines	90.25	0.80	-5.75

Nov 21st, 2022	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	664	-4	0
FMG Blended Fines	672	-4	8
Robe River	673	-4	9
Western Fines	675	-4	11
Atlas Fines	670	-4	6
Yandi	658	-4	-6

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	6.00	1.00
	High Grade Fe 63 - 64%	5.00	0.00
	High Grade Fe 64 - 65%	5.00	0.00
	High Grade Fe 65 - 65.5%	5.00	0.00
	Low Grade Fe	29.00	-1.00
1% Alumina	High Fe Grade Al <2.25%	59.00	5.00
	High Fe Grade Al 2.25-4%	5.00	0.00
	Low Fe Grade Al <2.25%	5.00	0.00
	Low Fe Grade Al 2.25-4%	14.00	-5.00
1% Silica	High Fe Grade Si <4%	15.00	-1.00
	High Fe Grade Si 4-6.5%	13.00	5.00
	Low Fe Grade	5.00	-3.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	5.00	0.00
	High Fe Grade 0.115%<P<0.15%	5.00	0.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	1.00	0.00
	High Grade Fe 63 - 64%	1.00	0.25
	High Grade Fe 64 - 65%	1.00	0.25
	High Grade Fe 65 - 65.5%	1.00	0.25
1% Alumina	High Fe Grade Al <2.25%	0.75	0.25
	High Fe Grade Al 2.25-4%	2.50	-0.25
1% Silica	High Fe Grade Si <4%	0.75	0.25
	High Fe Grade Si 4 - 6.5%	1.25	-0.25
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	3.00	0.25
	High Fe Grade 0.115%<P<0.15%	4.75	-0.25

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	0.00	Jingtang	-20.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-10.00	0.00

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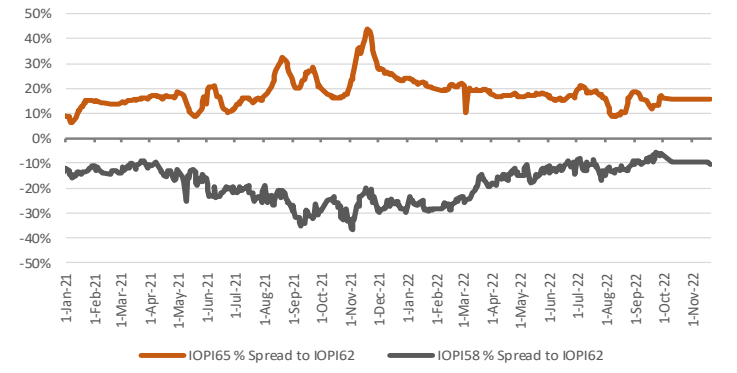
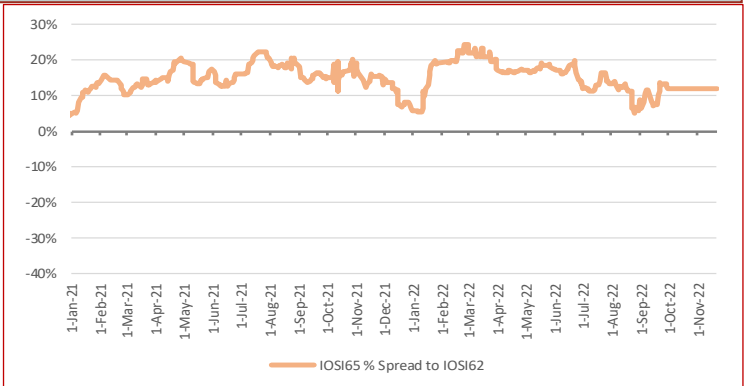
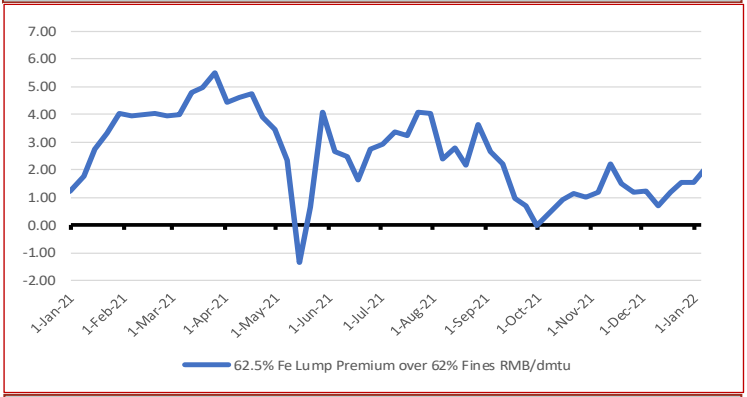
IRON ORE INDEX PREMIUMS/DISCOUNTS

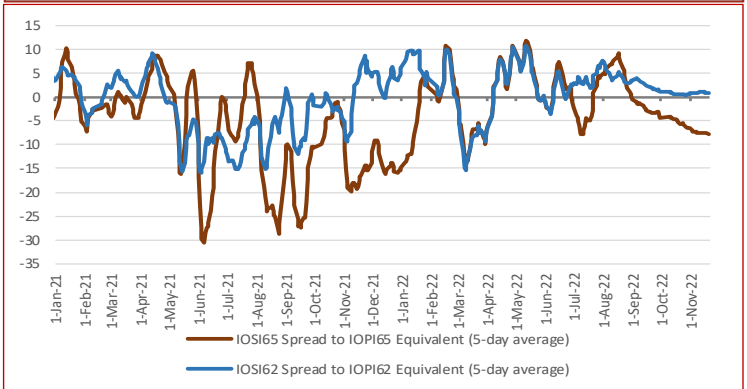




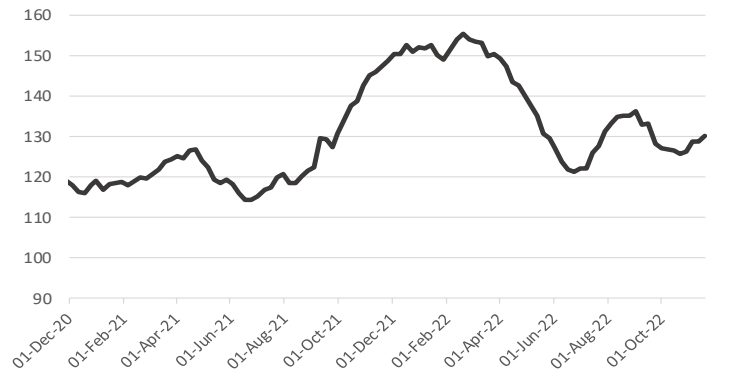
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



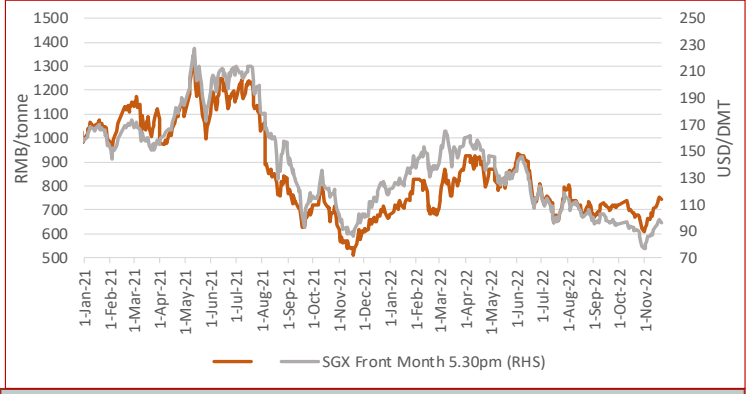
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

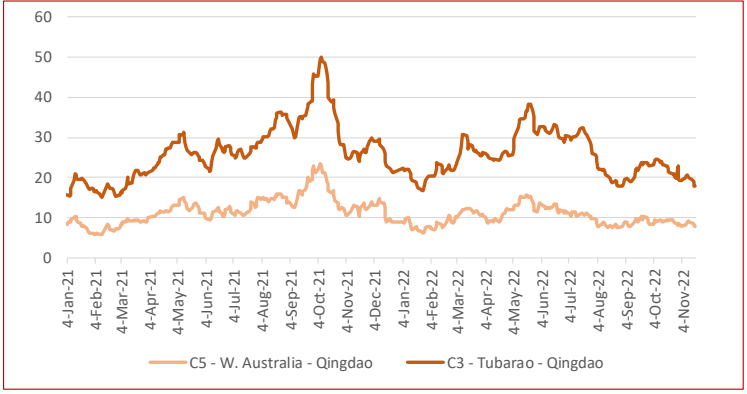
Week Ending Nov 18th, 2022

Province	This week	Change %	Low [2]	High [2]
Jingtang	13.10	1.55%	9.84	19.20
Qingdao	24.75	3.73%	9.41	26.24
Caofeidian	11.00	-6.78%	9.05	16.29
Tianjin	9.53	-6.84%	7.14	12.97
Rizhao	17.93	-0.28%	9.44	19.26
Total (35 Ports)	130.20	1.14%	98.80	155.39

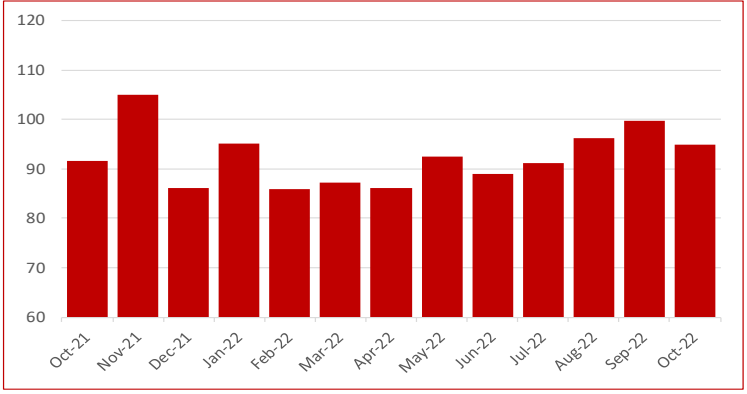
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Nov 21st, 3pm close			Nov 21st, 5:30pm		
Contract	I2301	Change	Change %	Dec. 22	Change	Change %
Closing Price	745.50	-8.00	-1.06%	95.80	-3.00	-3.04%
Vol traded ('000 lots)	98.98	-4.54	-4.38%	4.82	-1.29	-21.10%
Open positions ('000 lots)	76.71	-2.08	-2.64%	33.45	0.25	0.75%
Day Low	726.0	-7.50	-1.02%	94.35	-1.60	-1.67%
Day High	755.0	-2.50	-0.33%	98.90	-0.40	-0.40%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2022/11/18	Change	Change %
ReBar HRB400 ϕ18mm	3,840	50	1.32%
Wirerod Q300 ϕ6.5mm	4,340	70	1.64%
HRC Q235/SS400 5.5mm*1500*C	3,860	50	1.31%
CRC SPCC/ST12 1.0mm*1250*2500	4,370	0	0.00%
Medium & Heavy Plate Q235B 20mm	3,910	40	1.03%
GI ST02Z 1.0mm*1000*C	4,630	20	0.43%
Colour Coated Plate	7,500	0	0.00%

CHINESE STEEL MILL PROFITABILITY

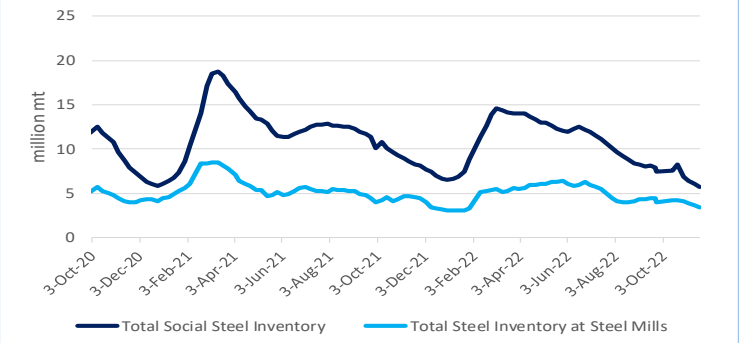
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	92.95	2	Mmi CFR Equivalent index for 1st Feb
Coke	2,740	0	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	2,910	100	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,459	78	Q234, incl. tax
Rebar cost - Blast furnace	3,942	88	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-122	-39	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	3,905	86	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	-35	-26	based on Shanghai prices, incl. tax

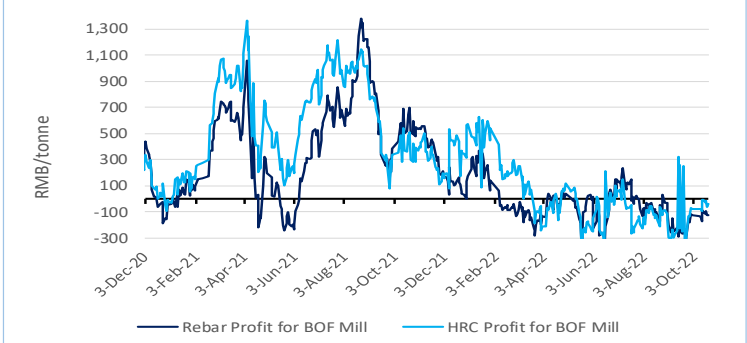
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

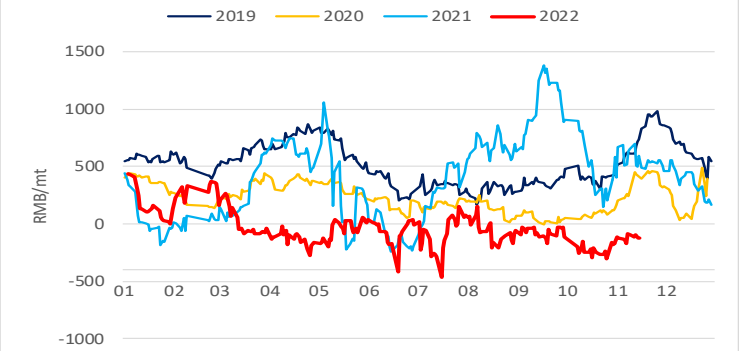
CHINESE STEEL INVENTORIES



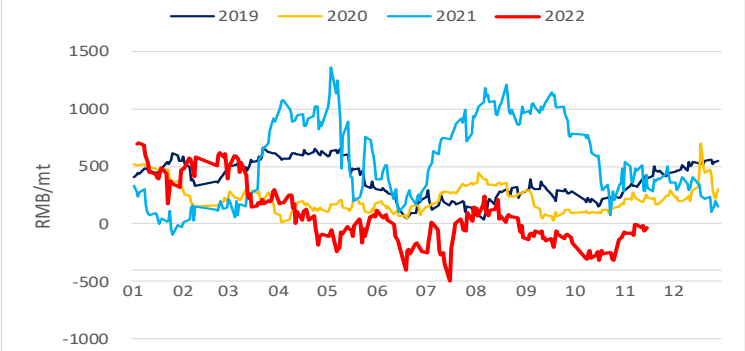
CHINESE STEEL MILL PROFITABILITY



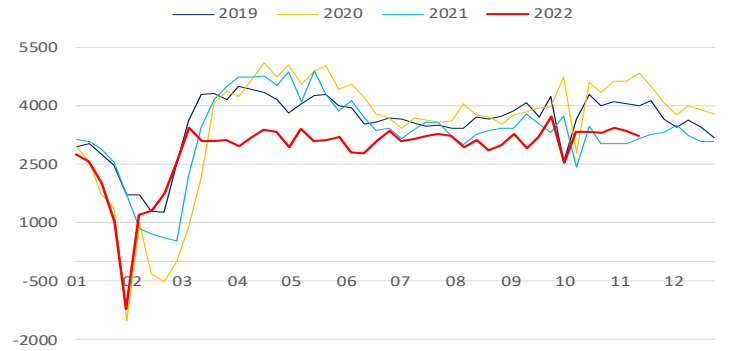
CHINESE STEEL MILL PROFITABILITY—Rebar



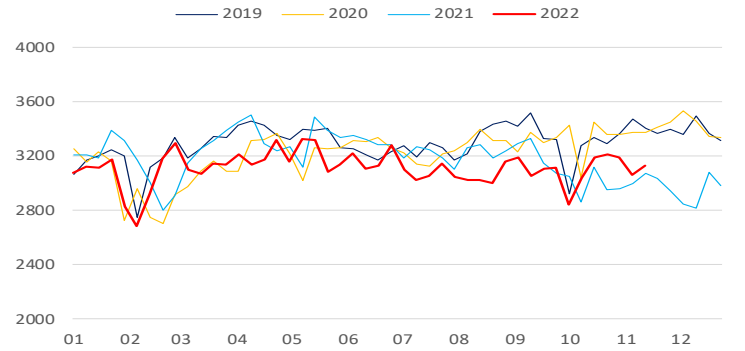
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



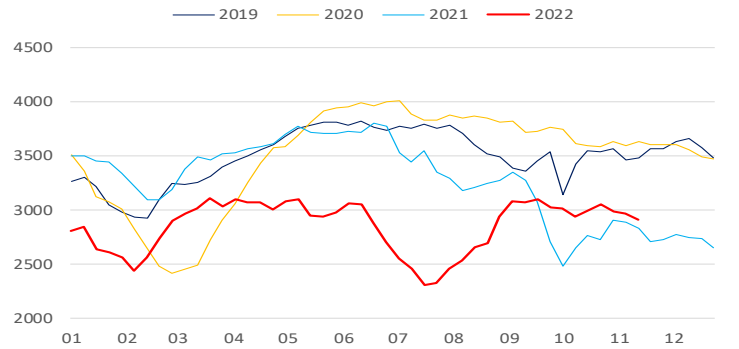
CHINESE STEEL CONSUMPTION—Rebar



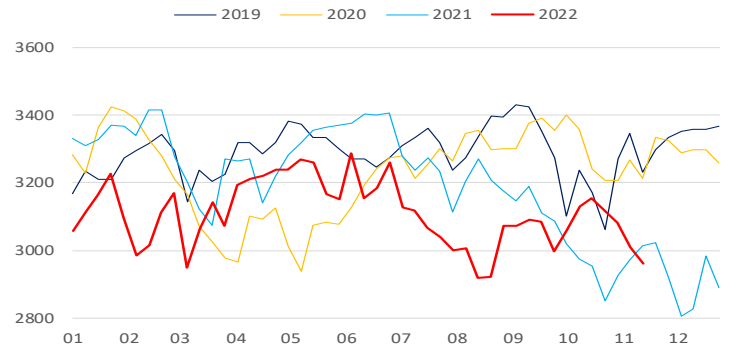
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Nov 21st, 2022	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Nov 21st, 2022	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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